

Integrating spatial and temporal metrics in smartphone-based finger tapping test: Insights into motor performance dynamics

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ARTICLE INFO

Keywords:

Finger Tapping Test
Smartphone
Motor performance
Fatigue
Digital health
Behavioral neuroscience

ABSTRACT

Background: The Finger Tapping Test (FTT) is a classical tool for assessing motor timing. While traditionally limited to temporal measures, smartphone-based versions enable simultaneous recording of both temporal and spatial parameters, opening new opportunities for digital phenotyping of motor behavior. We aimed to characterize the temporal evolution of both temporal and spatial metrics during a 30-second smartphone-based FTT. **New method:** Eighty healthy adults performed the FTT with both hands using a custom smartphone application. We quantified number of taps, intertap interval, total displacement, and elliptical area across six consecutive 5-second intervals. Friedman's ANOVA and Kendall's W were applied to assess time-dependent changes and effect sizes.

Results: Temporal parameters were sensitive to fatigue: tap count and total displacement decreased, while intertap interval increased across blocks. In contrast, the elliptical area remained stable, indicating preserved spatial control despite temporal deterioration. The largest effect size was found for intertap interval, particularly among female participants (Kendall's $W = 0.8$).

Comparison with existing methods: Unlike traditional FTT implementations that capture only timing, our smartphone-based approach integrates spatial metrics, revealing a dissociation between temporal fatigue and spatial stability during repetitive tapping.

Conclusion: These findings demonstrate the added value of spatial metrics in digital motor assessments, highlighting the potential of smartphone-based FTT as a scalable and low-cost tool for behavioral neuroscience. By bridging temporal and spatial domains, this approach advances methodological innovation in motor performance analysis and paves the way for applications in both research and clinical populations.

Introduction

The Finger Tapping Test (FTT) is a classical and widely adopted tool

for assessing motor control and neural conduction through rapid, repetitive finger movements (Jobbágy et al., 2005; Zhang et al., 2018). Traditionally implemented with mechanical or electronic devices such

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<https://doi.org/10.1016/j.jneumeth.2025.110645>

Received 16 August 2025; Received in revised form 21 October 2025; Accepted 28 November 2025

Available online 29 November 2025

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as telegraph keys or keyboards, the FTT has provided simple yet robust temporal measures of motor timing and coordination (Cousins et al., 1998; Hubel et al., 2013; Guo et al., 2022; Habets et al., 2023). These methods, however, have been inherently limited to capturing temporal parameters, offering only a partial view of motor performance.

The advent of smartphones and touchscreen technology has expanded the analytical potential of the FTT. Beyond precise recording of tapping times, smartphones enable the capture of spatial coordinates of each tap, thus permitting the simultaneous evaluation of temporal and spatial aspects of motor performance (Lee et al., 2016; Lipsmeier et al., 2009; Hitchcock et al., 2021; Broeder et al., 2023; Brito et al., 2023). This innovation aligns with the broader movement toward digital phenotyping and mobile health technologies, which emphasize scalable, low-cost, and ecologically valid behavioral assessments.

Among the spatial features available through touchscreen recordings, two stand out as particularly informative. Total displacement reflects the cumulative distance between consecutive taps, serving as an indicator of movement amplitude. Elliptical area, in turn, represents the region encompassing 95 % of tap positions, providing a proxy for spatial dispersion and motor stability (Lee et al., 2016; Brito et al., 2023; Lima et al., 2024). While temporal dynamics of FTT performance are well established, characteristically showing reduced tap frequency and increased intertap intervals as fatigue emerges (Hubel et al., 2013; Aydin et al., 2015; Heimhofer et al., 2024), the behavior of spatial metrics under fatigue conditions remains poorly understood.

Addressing this gap, the present study examined the temporal evolution of both spatial and temporal parameters during a 30-second smartphone-based FTT. We hypothesized that fatigue effects would manifest as increases in intertap interval and reductions in tap counts, but also as alterations in spatial performance, reflected by decreased total displacement and increased spatial dispersion. By integrating temporal and spatial analyses, our approach advances the methodological scope of the FTT and contributes to the development of innovative digital tools for behavioral neuroscience, in line with current efforts to modernize and expand the measurement of human behavior.

Methods

Ethical considerations

This study was approved by the Human Research Ethics Committee of the Tropical Medicine Centre of the Federal University of Pará (Report # 6.036.494). All participants provided written informed consent after receiving a detailed explanation of the experimental procedures.

Participants

In the present study, we recruited healthy individuals in order to establish a reference pattern of normality. Eighty healthy adults participated in the study (40 women: 34.6 ± 9.4 years; 40 men: 31.7 ± 8.0 years). None reported chronic or degenerative conditions that could affect motor performance. Handedness was assessed using the short version of the Edinburgh Handedness Inventory (Veale, 2014).

Procedures

A Samsung Galaxy S10 Plus smartphone was used to perform the FTT. The device featured a 6.4-inch Dynamic AMOLED QHD+ touchscreen (3040 × 1440 pixels; HDR10; Gorilla Glass 6; 536 ppi; 120 Hz touch sampling). The custom *Momentum Touch* application (Brito et al., 2023) was installed. This app was specifically designed for research purposes, enabling simultaneous acquisition of timestamped tap events, X–Y spatial coordinates, and Boolean flags for hits within the target area. Data were exported in open-format text files suitable for subsequent processing in Python.

During testing, participants were seated at a table with the smartphone placed in front of them. A central 20 mm square was displayed on the screen, serving as the tapping target to standardize performance.

Participants were instructed to tap inside the square as quickly and accurately as possible for 30 s using the index finger. Taps outside the square were recorded as errors. Each participant completed the test with both dominant and non-dominant hands, with the order randomized to minimize sequence and fatigue effects. A 1-minute rest interval was provided between trials, and participants were given a short familiarization period before formal testing.

Data analysis

The Momentum Touch app exported raw text files containing timestamp, X–Y coordinates, and hit/miss information for each tap. These files were processed using custom Python routines that implemented a reproducible analysis pipeline. The following metrics were extracted:

1. Number of taps: total count of valid taps.
2. Intertap interval: mean time between successive taps.
3. Total displacement: cumulative Euclidean distance between consecutive tap coordinates, reflecting movement amplitude.
4. Elliptical area: the area of an ellipse encompassing 95 % of taps, serving as a proxy for spatial dispersion.

To examine short-term temporal dynamics, all parameters were quantified in six consecutive 5-second intervals (0–5 s, 5–10 s, 10–15 s, 15–20 s, 20–25 s, and 25–30 s), consistent with previous studies (Heimhofer et al., 2024).

Statistical analysis

Muscle fatigue was inferred whenever a significant decline in Finger Tapping Test performance was observed across the analyzed intervals, indicated by an increase in inter-tap interval, a decrease in tap count, a reduction in movement amplitude, or an expansion of the ellipse area. Because the data did not meet normality assumptions, Friedman's ANOVA was applied to examine within-subject temporal effects for each variable and subgroup (sex × hand dominance). Post hoc pairwise comparisons were performed using Dunn's test with Holm correction for multiple comparisons. Effect sizes were computed using Kendall's W, interpreted according to Landis and Koch (1977) as slight ($0 \leq W < 0.20$), fair ($0.20 \leq W < 0.40$), moderate ($0.40 \leq W < 0.60$), substantial ($0.60 \leq W < 0.80$), and almost perfect ($W \geq 0.80$). All analyses and visualizations (boxplots and line graphs) were performed in Python, using the *scipy*, *pandas*, *statsmodels*, and *matplotlib* libraries.

Methodological framework Taken together, the smartphone-based tapping protocol, research-grade data acquisition app, and automated Python pipeline constitute a reproducible and low-cost methodological framework for motor behavior assessment. By enabling simultaneous analysis of temporal and spatial metrics, this approach advances the traditional scope of the FTT and provides a scalable tool for behavioral neuroscience research.

Results

This study examined time-dependent changes in four performance measures during a 30-second smartphone-based FTT, stratified by sex and hand dominance (Table 1). Significant differences ($p < 0.05$) were observed across blocks for number of taps and intertap interval (temporal metrics) and for total displacement (spatial amplitude), regardless of sex or hand. In contrast, elliptical area (spatial dispersion) showed minimal or no sensitivity to temporal change.

Effect sizes (Kendall's W) revealed distinct patterns. Intertap interval showed the strongest and most consistent effects, reaching almost perfect levels among females using the dominant hand ($W = 0.808$). Number of taps and total displacement presented moderate effects ($W \approx 0.30$ – 0.43 , depending on subgroup), while elliptical area displayed negligible values ($W \leq 0.11$); only the male dominant-hand group reached statistical significance with a small effect ($W = 0.107$).

Post-hoc comparisons confirmed progressive temporal deterioration

Table 1

Summary of Friedman ANOVA results and Kendall's W effect sizes for time-dependent variation in motor performance measures, stratified by sex and hand dominance.

Variable	Non-dominant hand			Dominant hand		
	χ^2	p-value	Effect size	χ^2	p-value	Effect size
Male						
Number of taps	70.78	0.0001	0.354	71.55	0.0001	0.358
Inter tap interval	104.69	0.0001	0.523	60.49	0.0001	0.302
Total displacement	74.13	0.0001	0.371	67.13	0.0001	0.336
Area	8.26	0.143	0.041	21.4	0.0001	0.107
Female						
Number of taps	86.99	0.0001	0.435	73.37	0.0001	0.367
Inter tap interval	151.81	0.0001	0.759	161.70	0.0001	0.808
Total displacement	74.37	0.0001	0.372	59.01	0.0001	0.295
Area	10.26	0.068	0.051	2.4	0.791	0.012

in tapping: number of taps declined and intertap interval increased across successive blocks, consistent with fatigue-related slowing. The spatial amplitude (total displacement) also decreased over time, particularly in later intervals, suggesting reduced movement extent as the task progressed. By contrast, spatial dispersion (elliptical area) remained essentially unchanged in nearly all comparisons.

Median and interquartile values by block are reported in [Table 2](#) (males) and [Table 3](#) (females). Full post-hoc results are provided in [Supplementary Material 1](#). [Fig. 1](#) illustrates the evolution of a temporal metric (intertap interval) and the spatial distribution of taps with the 95 % ellipse.

Summary of findings

Temporal metrics (tap count, intertap interval) and one spatial metric (total displacement) varied systematically across intervals, consistent with fatigue effects, whereas spatial dispersion (elliptical area) was stable, indicating a dissociation between temporal fatigability and spatial stability during repetitive tapping

Discussion

This study investigated the temporal evolution of both temporal and spatial parameters during a smartphone-based Finger Tapping Test. The

Table 2

Median values of performance parameters in the FTT across the six analyzed time intervals for male participants. Values in parentheses represent the interquartile range (IQR).

Intervals	Number of taps	Intertap interval (ms)	Total displacement (px)	Ellipse area (px ²)
Non-Dominant hand				
0–5 s	25 (5.5)	0.169 (0.02)	42439 (9590)	10500 (9571)
5–10 s	29.5 (4)	0.170 (0.02)	49849 (7513)	6703.6 (10977)
10–15 s	29 (4)	0.171 (0.02)	49273 (6727)	7126.2 (11850)
15–20 s	28 (4)	0.178 (0.02)	47402 (6796)	7203 (9856)
20–25 s	27 (4)	0.183 (0.02)	46648 (6855)	8805.6 (11463)
25–30 s	28 (3)	0.184 (0.02)	47710 (5096)	10492 (14069)
Dominant hand				
0–5 s	28 (5.3)	0.158 (0.03)	48132 (9282)	10260 (9245)
5–10 s	32 (4)	0.158 (0.02)	54199 (7683)	9835 (12469)
10–15 s	31 (4)	0.161 (0.02)	51827 (6217)	10178 (18000)
15–20 s	30 (3.3)	0.166 (0.02)	51162 (6863)	6892 (8040)
20–25 s	30 (3.3)	0.168 (0.02)	50591 (6541)	10394 (10431)
25–30 s	31 (2.8)	0.167 (0.02)	52833 (5825)	10815 (18577)

px: pixels; px²: pixel squared

Table 3

Median values of performance parameters in the FTT across the six analyzed time intervals for female participants. Values in parentheses represent the interquartile range (IQR).

Intervals	Number of taps	Intertap interval (ms)	Total displacement (px)	Ellipse area (px ²)
Non-Dominant hand				
0–5 s	24 (5.5)	0.180 (0.03)	40602 (9946)	11633 (17550)
5–10 s	26 (4)	0.190 (0.03)	44656 (6259)	10643 (12481)
10–15 s	26 (2.3)	0.196 (0.02)	43445 (4287)	8985 (10743)
15–20 s	24 (3)	0.206 (0.03)	41202 (6501)	10725 (12234)
20–25 s	24 (3.3)	0.209 (0.03)	40345 (5577)	10443 (15876)
25–30 s	25 (3)	0.212 (0.02)	41989 (5685)	11481 (13227)
Dominant hand				
0–5 s	27.5 (5)	0.170 (0.03)	46355 (8431)	10450 (9979)
5–10 s	29 (4)	0.174 (0.03)	48972 (6856)	10605 (9668)
10–15 s	27.5 (4)	0.181 (0.03)	46166 (6532)	10610 (10187)
15–20 s	27 (4.3)	0.188 (0.03)	44857 (7426)	10198 (9860)
20–25 s	26 (4)	0.191 (0.02)	43656 (7227)	10849 (11291)
25–30 s	27 (3)	0.194 (0.03)	45226 (5454)	10713 (10384)

main finding was a dissociation between temporal fatigability and spatial stability: while number of taps, intertap interval, and total displacement varied systematically across time blocks, the elliptical area remained unchanged. These results demonstrate that fatigue predominantly affects timing and movement amplitude, but not spatial dispersion, thereby expanding our understanding of motor performance dynamics in repetitive tapping tasks.

Consistent with previous studies, temporal performance deteriorated across intervals, characterized by reduced tap frequency and increased intertap intervals ([Tanner and Bowles, 1995](#); [Hubel et al., 2013](#); [Aydin et al., 2015](#); [Madinabeitia-Mancebo et al., 2021](#); [Heimhofer et al., 2024](#)). Among the temporal measures, the intertap interval proved to be the most sensitive indicator of fatigue, particularly in female participants, where effect sizes reached the almost perfect range. These results reinforce the role of intertap interval as a robust temporal marker of motor slowing in smartphone-based assessments.

Earlier work has described a polyphasic pattern during the initial seconds of tapping, with fluctuations in intertap interval within the first 6 s, followed by a gradual deceleration in later stages ([Tanner and Bowles, 1995](#)). A partial recovery of performance after approximately 15 s has also been reported, suggesting short-term adaptation during the task. In addition, [Heimhofer et al. \(2024\)](#) demonstrated that fatigue-related declines in tapping frequency are evident across age groups and fingers, confirming the generalizability of this phenomenon. In our study, we focused specifically on the index finger, the most commonly used in previous protocols, to ensure methodological consistency and comparability with the existing literature.

The incorporation of spatial metrics represents a methodological innovation made possible by touchscreen technology. Among these measures, total displacement decreased progressively across blocks, suggesting that fatigue not only slows motor execution but also reduces movement amplitude. In contrast, the elliptical area remained stable, indicating that participants preserved consistent spatial control despite temporal deterioration. This dissociation underscores the value of spatial dispersion as a complementary metric to distinguish fatigue-related changes in timing and amplitude from those associated with spatial variability.

In the present study, spatial variables exhibited distinct temporal profiles. Total displacement mirrored the trajectory of tap counts, with a peak observed between 5 and 10 s, followed by a gradual decline, whereas intertap interval followed an independent pattern of progressive slowing. Conversely, the elliptical area showed no significant variation across the test, reinforcing its robustness as a marker of spatial stability. An open methodological question remains regarding the effect of the central target square on participants' tapping behavior. It is

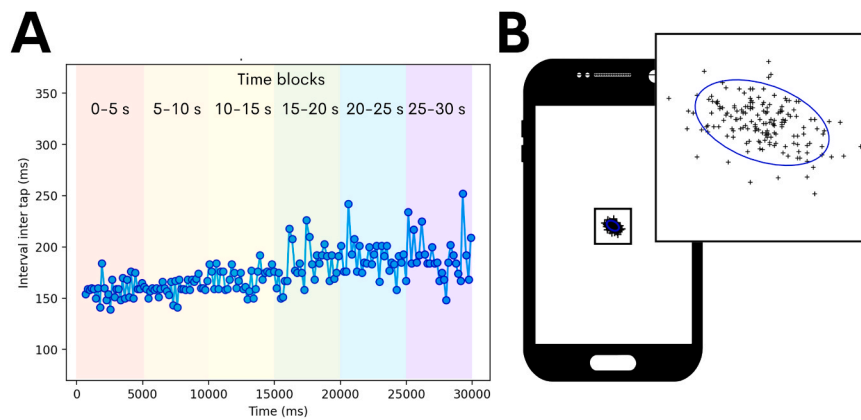


Fig. 1. Example of Finger Tapping Test (FTT) analysis. (A) Intertap interval plotted as a function of test duration, segmented into six time blocks: 0–5 s, 5–10 s, 10–15 s, 15–20 s, 20–25 s, and 25–30 s. (B) Spatial distribution of taps on the smartphone screen. The inset provides a zoomed-in view of the tapped area, along with an ellipse encompassing 95 % of the spatial coordinates.

possible that the visual constraint helped reduce spatial dispersion, and future investigations comparing conditions with and without the target area are needed to clarify this issue.

The analysis of spatial performance parameters in the FTT has only become feasible with the advent of touchscreen technology, making this dimension of motor assessment relatively underexplored. Evidence from previous studies indicates that spatial metrics are altered in clinical conditions such as Parkinson's disease (Lee et al., 2016; Broeder et al., 2023), underscoring their potential relevance for evaluating motor performance across diverse health contexts. Applying the present protocol to patients with disorders affecting neural conduction or motor coordination would help clarify whether the relationships between temporal, spatial, and global variables observed in healthy individuals are preserved or disrupted under pathological conditions.

Traditional FTT implementations have been restricted to temporal measures. By integrating both temporal and spatial metrics into a smartphone-based platform, this study establishes a novel methodological framework for digital motor assessments. The combination of a standardized tapping protocol, a research-grade mobile application, and an automated Python analysis pipeline ensures replicability, transparency, and scalability. This framework aligns with the growing emphasis on digital phenotyping in neuroscience and creates opportunities to assess motor behavior in both controlled laboratory settings and ecologically valid clinical environments.

Another key contribution of the present study is the analysis of results based on participants' sex and hand dominance. While some previous investigations (Hubel et al., 2013; Tanner and Bowles, 1995) have reported the performance of participants from both sexes using each hand to perform the FTT, other studies have specifically focused on male participants using their dominant hand to assess performance throughout the test (Aydin et al., 2015; Heimhofer et al., 2024).

The full potential and advantages of evaluating spatial parameters obtained via smartphones, and how these parameters relate to the assessment of muscle fatigue, remain unclear. The test duration is a factor that can be further explored to enhance the effects of fatigue on performance in the smartphone-based FTT. The 30-second duration adopted in this study was selected to ensure comparability with previous protocols and to minimize attentional or motivational decline in older participants. However, longer tapping periods (60–90 s), as proposed by Aydin et al. (2015) and Arias et al. (2012), may further amplify fatigue effects, especially in the inter-tap interval, which exhibited the strongest effect sizes here, and should be investigated in future work.

Another limitation of the present study is that testing was conducted exclusively on healthy young adult participants. To fully assess the predictive value of fatigue evaluation during the FTT, it is essential to extend this research to include older adults or patients with neurological

conditions that impact muscle strength or motor coordination. Several studies have shown that the relative reduction in performance during motor tasks is generally smaller in older adults than in young individuals, indicating greater resistance to muscle fatigue under certain effort conditions (Christie et al., 2011). This pattern suggests that, during the Finger Tapping Test, older adults may also exhibit an attenuation in the decline of tapping speed over time. Quantifying this temporal dynamic may represent a potential digital biomarker of healthy motor aging, reflecting the ability to maintain performance in repetitive rhythmic tasks. In patients with neurological disorders, when applying a test such as the Finger Tapping Test (rapid repetitive tapping), one can expect a greater decline in tapping rate over time, increased inter-tap variability, and/or reduced consistency compared with healthy controls or age-matched individuals. This pattern is consistent with findings from several studies showing that muscle fatigue is a prominent feature in conditions such as multiple sclerosis and Parkinson's disease (Royer et al., 2022; Kluger et al., 2013; Patejdl and Zettl, 2022). Our group intend to extend this kind of approach to elderly people and patients with neurological disorders to test these hypotheses.

Nevertheless, there is also a pressing need to establish normative data in healthy populations, as well as efforts to standardize methods for evaluating motor performance using the smartphone-based FTT. The dissociation observed here has important implications. First, it suggests that spatial dispersion is resilient to short-term fatigue, which may provide a stable marker for distinguishing pathological motor variability from fatigue-related changes. Second, the accessibility and low cost of smartphone-based testing make it feasible for large-scale behavioral studies and remote monitoring. Future work should extend this protocol to older adults and patient populations, such as those with Parkinson's disease or peripheral neuropathies, to determine whether similar dissociations are preserved or altered under pathological conditions. Moreover, integration with machine learning approaches could enable automated detection of motor deterioration, expanding the clinical utility of the method.

By disentangling temporal fatigability from spatial stability, our approach not only advances the methodological scope of the FTT but also contributes to the development of scalable digital biomarkers for behavioral neuroscience and clinical neurology.

CRediT authorship contribution statement

Bianca Callegari: Writing – review & editing, Funding acquisition. **Luis Carlos Pereira Monteiro:** Writing – review & editing, Formal analysis. **Anselmo de Athayde Costa e Silva:** Writing – review & editing. **André dos Santos Cabral:** Writing – review & editing. **Cândida Helena Lopes Alves:** Writing – review & editing. **Alex Crisp:** Writing –

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. All authors have read and approved the final version of the manuscript and agree with its submission.

Acknowledgements

This research was funded by FAPESPA and CNPQ fundings. GS and BC are CNPq Fellows and receive CNPq Productivity Grant (#309936/2022–5 and #304102/2024–5). The funders had no role in the study design.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jneumeth.2025.110645](https://doi.org/10.1016/j.jneumeth.2025.110645).

Data Availability

Data will be made available on request.

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